HW9 Exercise I: K-Mean (6710503739)

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1 Exercise I: K-Mean

Points A1–A9, K = 3, Euclidean distance, initial seeds A1, A4, A7.

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

points = {"A1": (3, 8), "A2": (9, 4), "A3": (4, 9), "A4": (2, 2), "A5": (10, 5),
          "A6": (2, 4), "A7": (6, 8), "A8": (4, 3), "A9": (4, 7)}
names = list(points)
X = np.array(list(points.values()), dtype=float)
centroids = X[[0, 3, 6]]
colors = ["tab:red", "tab:green", "tab:blue"]
```

[2]:
```
iteration = 0
while True:
    iteration += 1
    dist2 = ((X[:, None, :] - centroids) ** 2).sum(axis=2)
    labels = dist2.argmin(axis=1)
    new_centroids = np.array([X[labels == k].mean(axis=0) for k in range(3)])

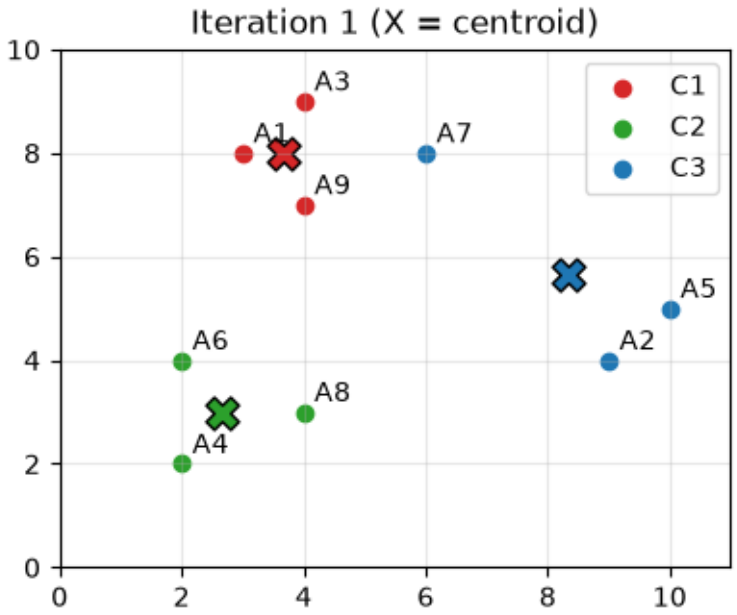
    table = pd.DataFrame(dist2, index=names, columns=[f"dist2 to C{k+1}" for k in␣
 ↪range(3)]).round(2)
    table["cluster"] = [f"C{l+1}" for l in labels]
    print(f"Iteration {iteration}  (centroids used: {[tuple(c.round(2).tolist())␣
 ↪for c in centroids]})")
    print(table.to_string())
    for k in range(3):
        members = [n for n, l in zip(names, labels) if l == k]
        print(f"  C{k+1}: members = {members}, new centroid =␣
 ↪{tuple(new_centroids[k].round(2).tolist())}")

    fig, ax = plt.subplots(figsize=(4.5, 3.5))
    for k in range(3):
        ax.scatter(*X[labels == k].T, c=colors[k], label=f"C{k+1}")
        ax.scatter(*new_centroids[k], c=colors[k], marker="X", s=150,␣
 ↪edgecolors="k")
```

```
    for n, (x, y) in points.items():
        ax.annotate(n, (x, y), xytext=(4, 4), textcoords="offset points")
    ax.set_title(f"Iteration {iteration} (X = centroid)")
    ax.set_xlim(0, 11); ax.set_ylim(0, 10); ax.legend(); ax.grid(alpha=0.3)
    plt.show()

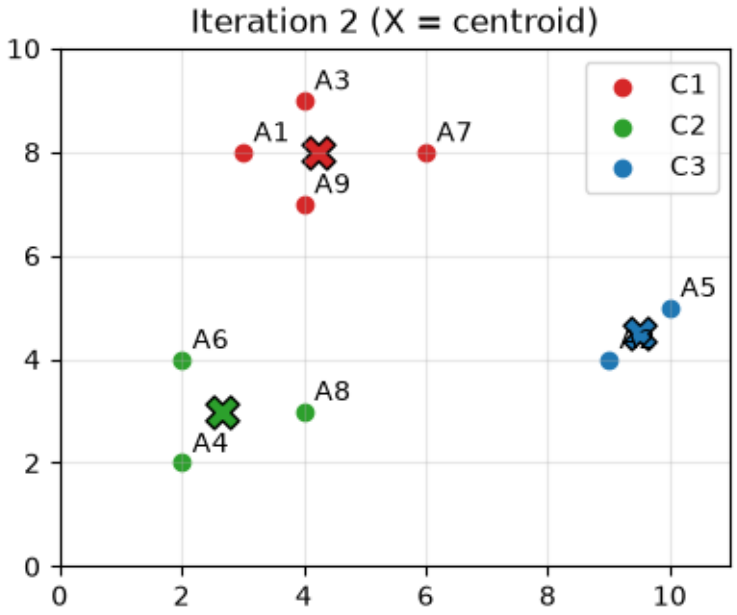
    if np.allclose(new_centroids, centroids):
        break
    centroids = new_centroids
print(f"Converged after {iteration} iterations")
```

```
Iteration 1  (centroids used: [(3.0, 8.0), (2.0, 2.0), (6.0, 8.0)])
    dist2 to C1  dist2 to C2  dist2 to C3 cluster
A1          0.0         37.0          9.0      C1
A2         52.0         53.0         25.0      C3
A3          2.0         53.0          5.0      C1
A4         37.0          0.0         52.0      C2
A5         58.0         73.0         25.0      C3
A6         17.0          4.0         32.0      C2
A7          9.0         52.0          0.0      C3
A8         26.0          5.0         29.0      C2
A9          2.0         29.0          5.0      C1
  C1: members = ['A1', 'A3', 'A9'], new centroid = (3.67, 8.0)
  C2: members = ['A4', 'A6', 'A8'], new centroid = (2.67, 3.0)
  C3: members = ['A2', 'A5', 'A7'], new centroid = (8.33, 5.67)
```

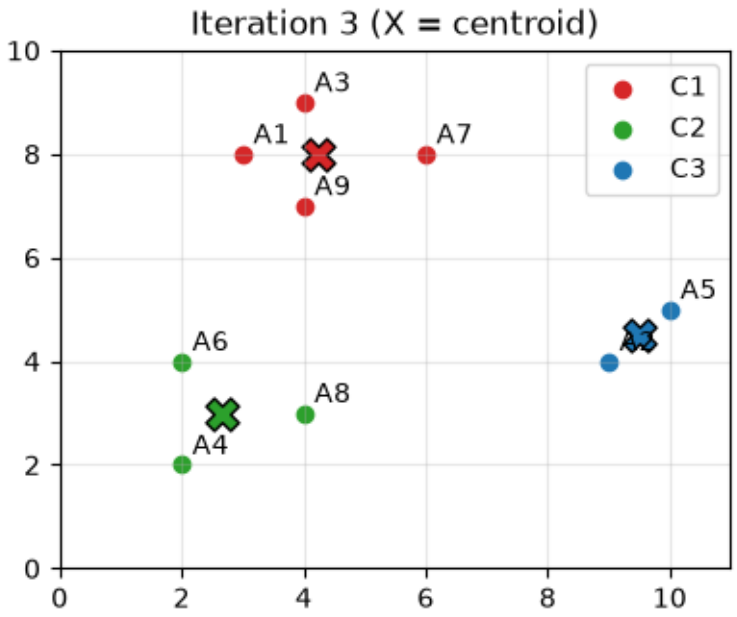



```
Iteration 2  (centroids used: [(3.67, 8.0), (2.67, 3.0), (8.33, 5.67)])
    dist2 to C1  dist2 to C2  dist2 to C3 cluster
```

```
A1          0.44         25.11         33.89       C1
A2         44.44         41.11          3.22       C3
A3          1.11         37.78         29.89       C1
A4         38.78          1.44         53.56       C2
A5         49.11         57.78          3.22       C3
A6         18.78          1.44         42.89       C2
A7          5.44         36.11         10.89       C1
A8         25.11          1.78         25.89       C2
A9          1.11         17.78         20.56       C1
  C1: members = ['A1', 'A3', 'A7', 'A9'], new centroid = (4.25, 8.0)
  C2: members = ['A4', 'A6', 'A8'], new centroid = (2.67, 3.0)
  C3: members = ['A2', 'A5'], new centroid = (9.5, 4.5)
```




```
Iteration 3  (centroids used: [(4.25, 8.0), (2.67, 3.0), (9.5, 4.5)])
    dist2 to C1  dist2 to C2  dist2 to C3 cluster
A1          1.56         25.11          54.5       C1
A2         38.56         41.11           0.5       C3
A3          1.06         37.78          50.5       C1
A4         41.06          1.44          62.5       C2
A5         42.06         57.78           0.5       C3
A6         21.06          1.44          56.5       C2
A7          3.06         36.11          24.5       C1
A8         25.06          1.78          32.5       C2
A9          1.06         17.78          36.5       C1
  C1: members = ['A1', 'A3', 'A7', 'A9'], new centroid = (4.25, 8.0)
  C2: members = ['A4', 'A6', 'A8'], new centroid = (2.67, 3.0)
  C3: members = ['A2', 'A5'], new centroid = (9.5, 4.5)
```




```
Converged after 3 iterations
```

1.1 Answer

Iteration	C1 members / centroid	C2 members / centroid	C3 members / centroid
1	A1, A3, A9 / (3.67, 8)	A4, A6, A8 / (2.67, 3)	A2, A5, A7 / (8.33, 5.67)
2	A1, A3, A7, A9 / (4.25, 8)	A4, A6, A8 / (2.67, 3)	A2, A5 / (9.5, 4.5)
3	A1, A3, A7, A9 / (4.25, 8)	A4, A6, A8 / (2.67, 3)	A2, A5 / (9.5, 4.5)

K-Means converges after 3 iterations (iteration 3 gives no change in members or centroids).

Final clusters: C1 = {A1, A3, A7, A9}, centroid (4.25, 8); C2 = {A4, A6, A8}, centroid (2.67, 3); C3 = {A2, A5}, centroid (9.5, 4.5).